

Statistical Data Analysis

Lecture 7: First introduction to nonparemetric methods

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Periods 4 & 5

⏮ Chapter 5: The bootstrap

📍 Chapter 6: Nonparametric methods

- Section 6.1 – The one-sample problem
- ▶▶ Section 6.2 – Asymptotic efficiency
- ▶▶ Section 6.3 – Two-sample problems
- ▶▶ Section 6.4 – Tests for correlation

Nonparametric Tests

Nonparametric Tests – Idea

- › Recall concepts
 - › Actual level (size) of test: (supremum of) probability of type I error
 - › A test has significance level α if its size is $\leq \alpha$
 - › Having the actual level larger than the pre-set α is undesirable
 - › Power of a test: probability of correctly rejecting the null

One-Sample t -Test: Parametric Location Test

- › The one-sample t -test is used to test hypotheses about the population mean
 - › Assumption $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} F = N(\mu, \sigma^2)$, with $\sigma > 0$
 - › Null hypothesis $H_0 : \mu = \mu_0$
 - › Test statistic $T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$.
 - › Null distribution: under H_0 , $T \sim t_{n-1}$
 - › Critical region: Reject if $|T| > t_{n-1, \alpha/2}$
- › Problem: if F not normal $\rightsquigarrow$ t approximation flawed
 - › Actual level of t -test in this case?
 - › If $T \sim t_{n-1}$, $P(|T| > t_{n-1, \alpha/2}) \neq \alpha$
 - › Actual power of t -test in this case?

Nonparametric Tests – Idea

- **Problem:** using parametric tests when distributional assumptions **do not** hold $\rightsquigarrow$ **actual** test level and power **deviate** from theoretical ones
- **Remedy:** nonparametric (i.e., **distribution-free**) tests
 - No parametric assumptions about underlying distribution F
 - **Actual** level is α for a broad class of distributions
 - Test statistics **nonparametric** under the null $\rightsquigarrow$ null distribution does not depend on F
 - Greater **power** than parametric tests when their distributional assumption do not hold
 - **Caveat:** parametric tests have greater power when their distributional assumptions hold

t-Test when $F \neq N$ – Illustration

```
# Want to test H_0: mu = 1 (mu_zero is 1), use significance level alpha = 5%
n <- 20
reps <- 10^3
pval_norm <- numeric(reps); pval_exp <- numeric(reps)
set.seed(123)
for (i in 1:reps){
  # Simulate data from normal with mean 1 and test, extract p_value
  pval_norm[i] <- t.test(rnorm(n, mean = 1, sd = 1), mu = 1)$p.value
  # Simulate data from exponential with mean 1 and test, extract p_value
  # Recall that for exponential distribution, mean = 1/rate
  pval_exp[i] <- t.test(rexp(n, rate = 1), mu = 1)$p.value
}

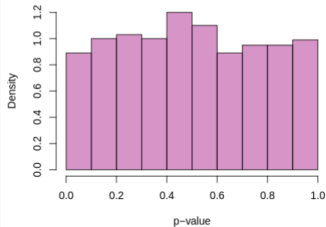
# Calculate frequency of type I error (i.e., mistakenly rejecting H_0 when true)
paste0("Type I error frequency for N(1, 1) samples: ", sum(pval_norm < 0.05)/reps)

## [1] "Type I error frequency for N(1, 1) samples: 0.044"

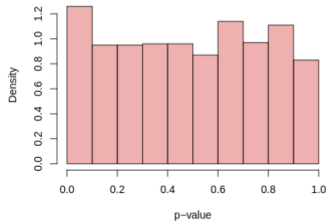
paste0("Type I error frequency for Exp(1) samples: ", sum(pval_exp < 0.05)/reps)

## [1] "Type I error frequency for Exp(1) samples: 0.08"
```

p-values for N(1, 1) samples



p-values for Exp(1) samples



Summary and wrap-up

- Parametric tests may fail if parametric assumptions are not satisfied
- Probability of Type I error may lie above α

Wrap-up

- › What now?
 - › Continue working on Assignment 4
 - › Deadline: 31 March 23:59
- › Next week no lecture or tutorials
 - › After that, lectures are on Wednesdays **at 11:00**
- › Good luck with your exams!